

THE PATH TO 2050

The proposal argues for an engine. This portfolio opens it: what it is made of, what it already found, which tools build it, what is still missing, and how anyone can rerun every number.

Arshad Akhtar Abbasia · applicant, PhD in Logistics, Molde University College
Code: github.com/Asthenosphere/computational_Logistics · Proposal: five pages

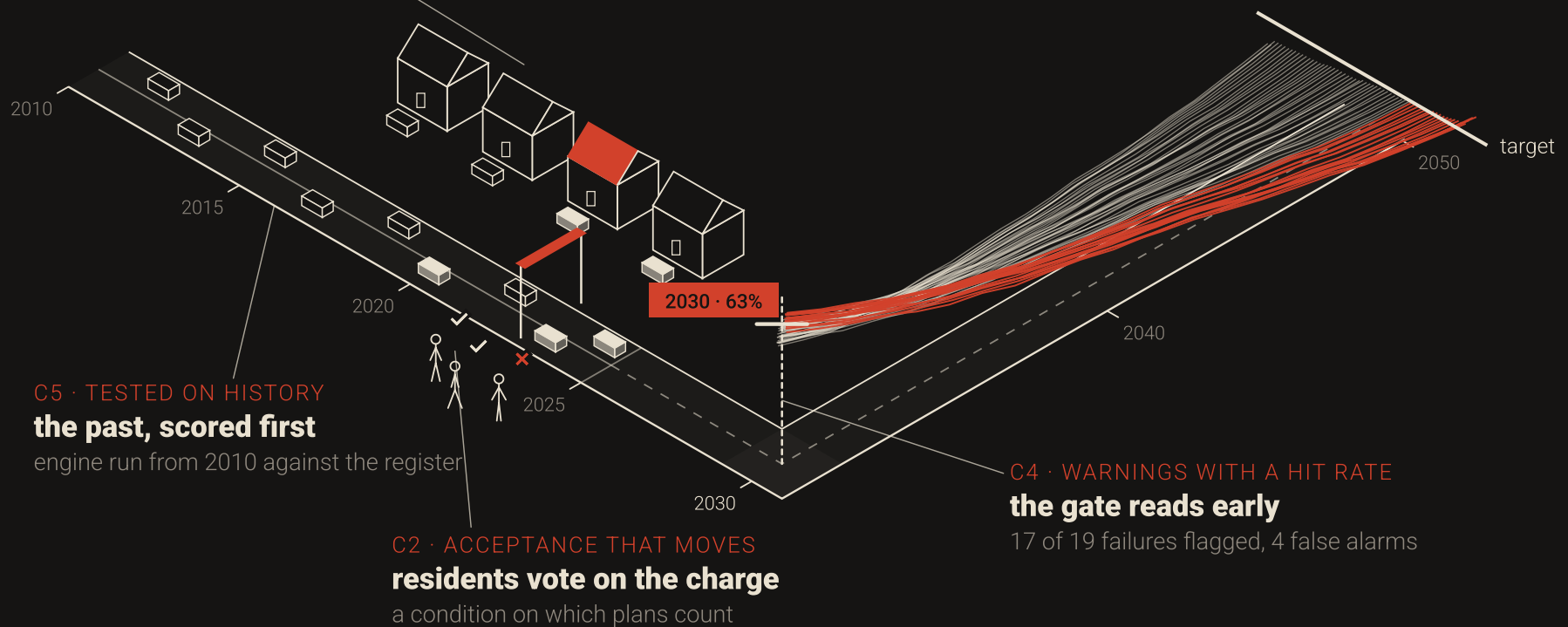
the whole project, on one road

each household turns its own car

replacement year by year, not an end state

sixty futures after the gate

preferences, chance and the future, split apart



THE ROAD SPLITS AT 2030

how to read this

EACH CHAPTER ANSWERS A QUESTION

A proposal has five pages and cannot show its workings. These chapters are the workings. Each one starts from a question a careful

reader would ask, and answers it with something built and run rather than promised.

CHAPTER	THE QUESTION IT ANSWERS	WHO ASKS IT FIRST
1 · The stock problem	Why households, and why a path rather than an end state?	TRANSPLAN, research area 1
2 · Five contributions in depth	What exactly is new, against what, and how will each claim be tested?	Every examiner
3 · Two towns and their networks	Why Molde and Kristiansund, and is this a logistics thesis?	The PhD committee
4 · Inside the engine	What are the equations, and do the estimators recover what they should?	Choice modellers
5 · What the prototype found	Does the chain produce decisions, and what does robustness cost?	Operations research
6 · The toolbench	Which tools exist, which are tested, which are still missing?	Anyone judging feasibility
7 · Rerun it yourself	Can I trust a number without trusting the author?	Anyone
8 · Three years, and the limits	Can one person finish it, and what does it not do?	The committee

THE STOCK PROBLEM

Norway won the showroom. The street has not caught up.

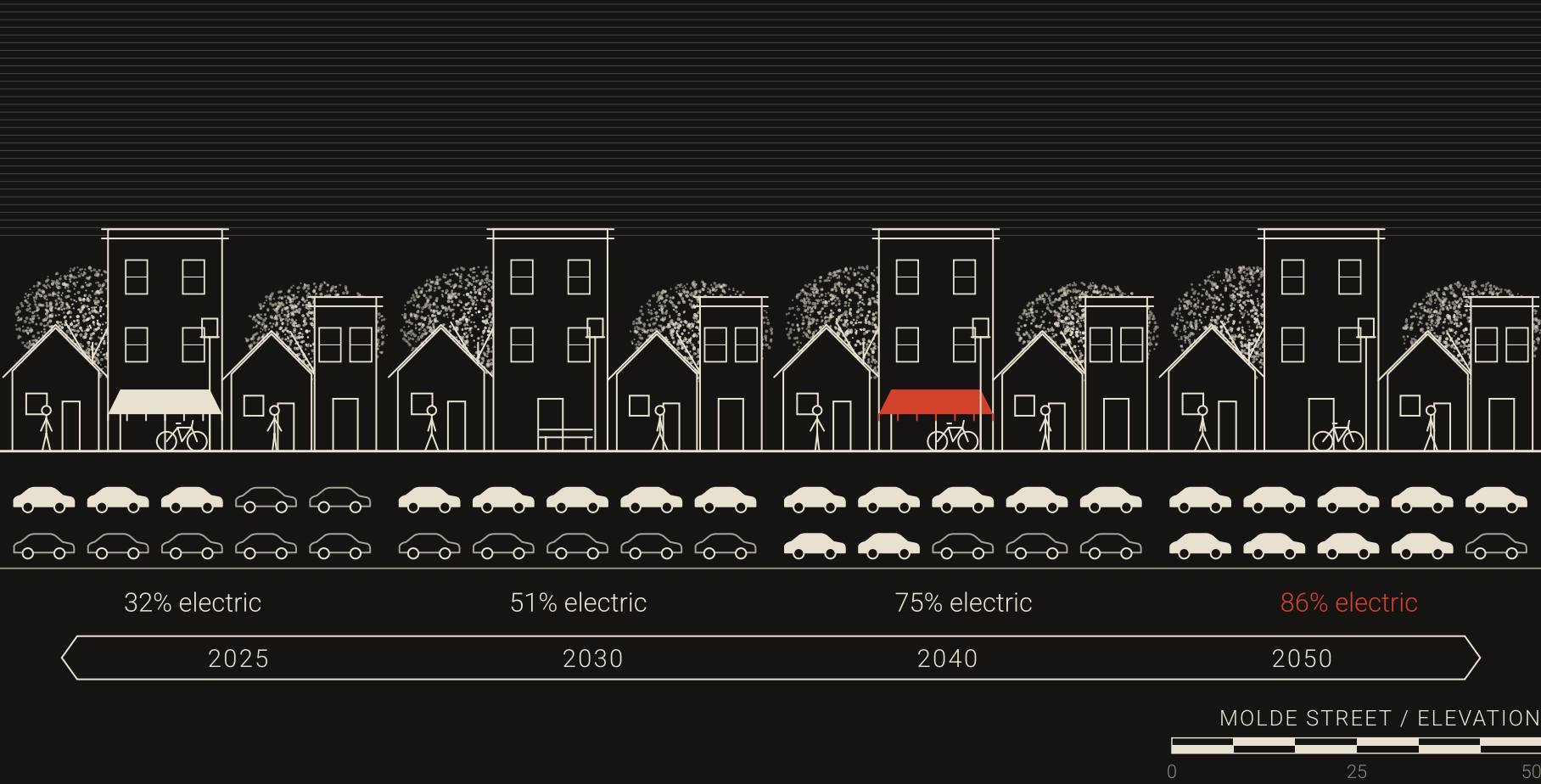
Picture one household in Molde in 2025. They drive a 2014 diesel that still runs. The new car they will buy is almost certainly electric: 95.9% of new cars sold in Norway in 2025 were¹. But they will buy it when the diesel dies or the family changes, not when a national plan would like them to. Multiply that household by every driveway in the country and you have the reason the fleet lags the showroom by well over a decade.

That lag is the part of the transition national models do not describe. TØI's review of models for backcasting says RTM and NTM6 compute

partial equilibria but “say nothing about the path to a new equilibrium”². The National Transport Plan asks for methods that can steer toward 2050³, and the methods assignment admits today's models suit continued policy, not deep cuts⁴. A backcast that cannot see the path cannot tell a planner whether 2030 is on track.

The path also carries the politics. As the fleet turns electric, fuel tax revenue falls, and the distance-based charge meant to replace it has stalled between the state and the municipalities⁵. Who accepts a charge depends on the car they own and the alternatives they have. So the same household that decides the fleet also decides whether the policy survives. That is why the engine is built from households.

the stock problem, in one street



THE STOCK TURNS

one street, four moments. each lit car is electric, in the share the prototype engine simulates under the most robust package. the street changes only as fast as households replace the cars they already own.

Prototype, median future, package P12; without policy the 2050 share is 74%. The prototype's purchase model is not yet calibrated to Norwegian sales and turns the fleet more slowly than TØI's fleet model BIG⁶; calibrating it to the vehicle register is the first job of article 2. The plate shows the mechanism, not the expected fleet.

FIVE CONTRIBUTIONS IN DEPTH

The proposal states each contribution in a sentence. Here each one gets the same five lines: what is new, the nearest work, what the

prototype already shows, what the article adds, and how the claim will be tested. If a line is weak, it is visible.

C1 · THE PATH A household engine that shows how Norway gets from today to 2050, year by year NEW Fleet turnover, mode choice and acceptance carried by the same households, so the transition path and its politics come out of one model. NEAREST BIG follows vintages with an estimated choice model, in aggregate and for fixed scenarios⁶. Deuten et al. use system dynamics for the Norwegian EV transition⁷. PROTOTYPE 1,000 agents in three zones, a yearly replacement and EV choice, mode choice with congestion, welfare and public money per household (chapter 4). ARTICLE A2 replaces synthetic preferences with A1’s estimates, gives replacement a memory of car age, and runs on the real networks of both towns. TEST Several observed patterns at once: travel survey mode shares, fleet age and fuel from the register, traffic counts⁸.	C2 · ACCEPTANCE THAT MOVES Acceptance measured in choice data, shifted as a charge is lived, and used as a condition on which plans count NEW Acceptance enters the robust search as a constraint, not a footnote, and it changes with experience rather than staying at its survey value. NEAREST Le Pira, Marcucci and Gatta gave freight agents acceptance from choice models⁹. TØI’s national experiment measured support for distance charges¹⁰. Support rose after Stockholm’s charge¹¹ but moved little in the Oslo field trial¹⁰. PROTOTYPE A self-interest proxy: the share of commuters not made worse off in 2030. It is a placeholder and is labelled as one. ARTICLE A1 measures acceptance at one point in time, with trust measured by attitudinal indicators in a hybrid choice model. How far acceptance shifts after introduction is set from the field evidence and every result is reported with and without the shift, because the evidence points both ways. TEST National attribute ranges, so Molde and Kristiansund can be compared with the national result directly.
C3 · HOW MUCH WE DO NOT KNOW 2050 uncertainty split into three sources, and a test of whether the winning plan survives each NEW Error in estimated preferences, chance inside the simulation and deep uncertainty about the future are separated over a 25-year policy path. NEAREST Petrik and colleagues propagated parameter error through one simulated day¹². Tveter, Welde and Odeck quantify uncertainty in the appraisal of single road projects¹³; I separate its sources along a policy path. PROTOTYPE Deep uncertainty only: 60 futures over nine uncertain inputs. Every package uses the same random draw, so chance is not yet separated. ARTICLE A2 draws preferences from their estimated covariance and repeats runs, so each source gets its share of the 2050 spread. TEST Does the ranking of packages change when each source is switched off in turn.	C4 · WARNINGS WITH A HIT RATE Pathways stress-tested across thousands of futures, with signposts whose lead time and false-alarm rate are measured NEW Each signpost carries its years of warning, false alarms and missed failures, as in adaptive water planning, applied to transport backcasting. NEAREST Raso et al. and Haasnoot et al. measure signals in water management^{14,15}; transport backcasting remains national and deterministic¹⁶. PROTOTYPE A 2030 reading of 63% of 2025 emissions flags 17 of 19 later failures with 4 false alarms. The threshold was fitted on the same 60 futures. ARTICLE A3 chooses thresholds on one set of futures and scores them on a held-out set, for several indicators planners can read. TEST Hit rate and false alarms out of sample, and lead time in years.

The engine is checked against Norway’s 2010 to 2025 electric transition before it is trusted with 2050

NEW	A household engine scored against a real transition before it is used for a future one.
NEAREST	Deuten et al. tested system dynamics on 2010 to 2017 ⁷ . EV forecasts often fail against what happened ¹⁷ , and Norwegian fixed-link forecasts underestimated long-run traffic, in some cases by half ¹⁸ .
PROTOTYPE	Not built. It is one validation step of A2, not an article.
ARTICLE	The 2010 run holds the 2026 preferences fixed, so it scores the fleet mechanics, not the stability of preferences, and is reported on that basis.
TEST	Simulated against registered EV fleet share by year, from register extracts at zone level.

TWO TOWNS AND THEIR NETWORKS

Molde and Kristiansund are case studies, not a national sample. They are car-dependent, the instruments that matter (tolls, bus service, parking, cycling) are decided locally, and their road networks

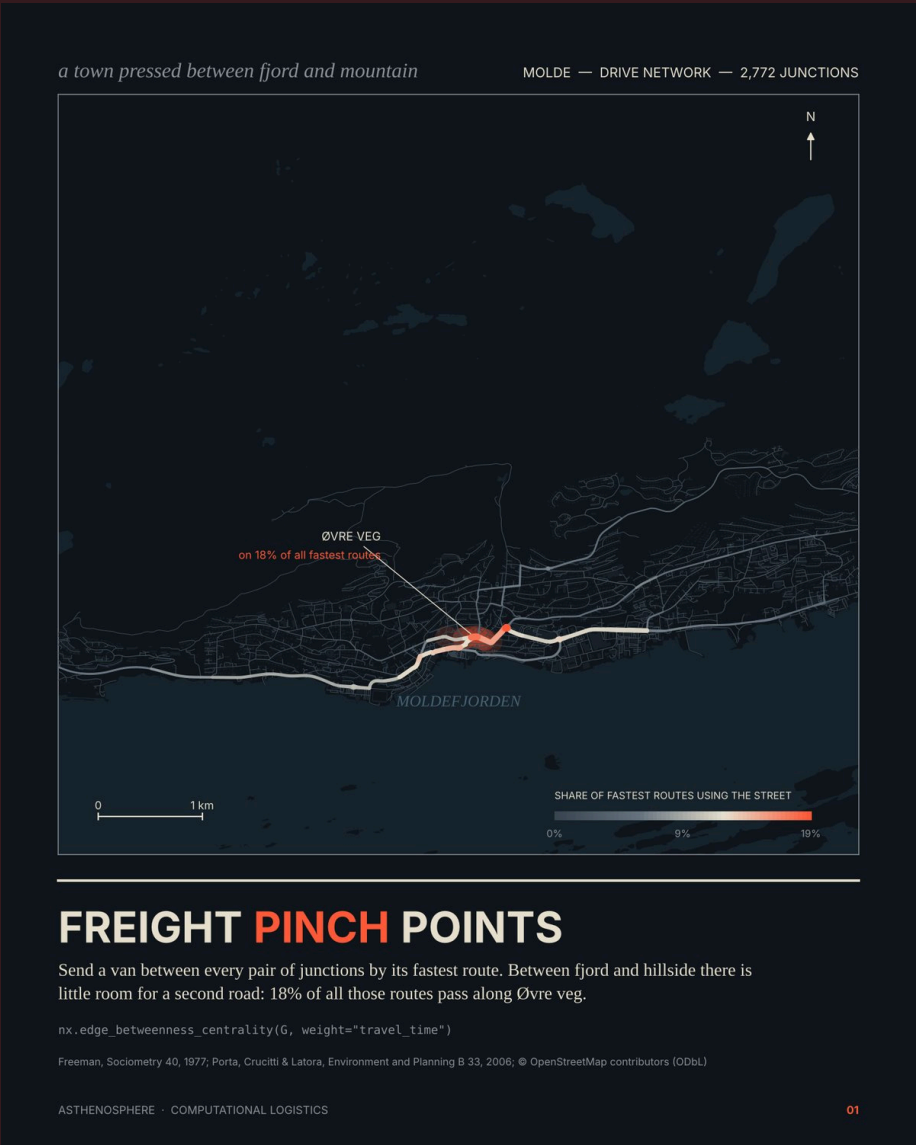
concentrate travel on a few links, which makes policy effects visible. Molde is the main case where preferences are estimated. Kristiansund is the transfer case, where the Molde model is applied without re-estimation.



The same network already answers logistics questions

The road graph that carries commuters in the engine is the one a delivery van drives. I built four plates on it for Molde to show the tools are not tied to commuting. Read in order, they go from where the

network is fragile, to what breaking it costs, to where to stand to serve the town, to how to drive it once you stand there. This is the bridge to article 3's freight test and to the logistics programme the PhD sits in.



01 · Freight pinch points. Route a van between every pair of junctions and shade each street by the share of routes on it. Øvre veg carries 18%.



02 · Where the detour goes. Close that street and route all 7.7 million junction pairs again: 32% of trips slower, by 10 s on average and 2 minutes at worst; 0.9% become impossible.



03 · The five-minute depot. Time the drive from every junction to every other and keep the best: 79% of buildings within five minutes, against 56% from the median junction.



04 · The shortest loop. One van, 24 addresses, a solver orders them: 48 minutes of driving instead of 137. OR-Tools agrees with PyVRP within 0.1%.

Free-flow drive times from OpenStreetMap speeds; no turn penalties, signals or congestion; flat terrain. Scripts in `skills/visual-`

narrative/scripts.

INSIDE THE ENGINE

Four layers, each built by one article on top of the one below. This chapter writes down what each layer computes in the prototype, and

what changes when the article replaces it.

LAYER 3

futures

10⁴ worlds every plan
is tested against

A3

LAYER 2

households

cars, trips and opinions
carried through time

A2

LAYER 1

network

the real streets
of molde, from osm

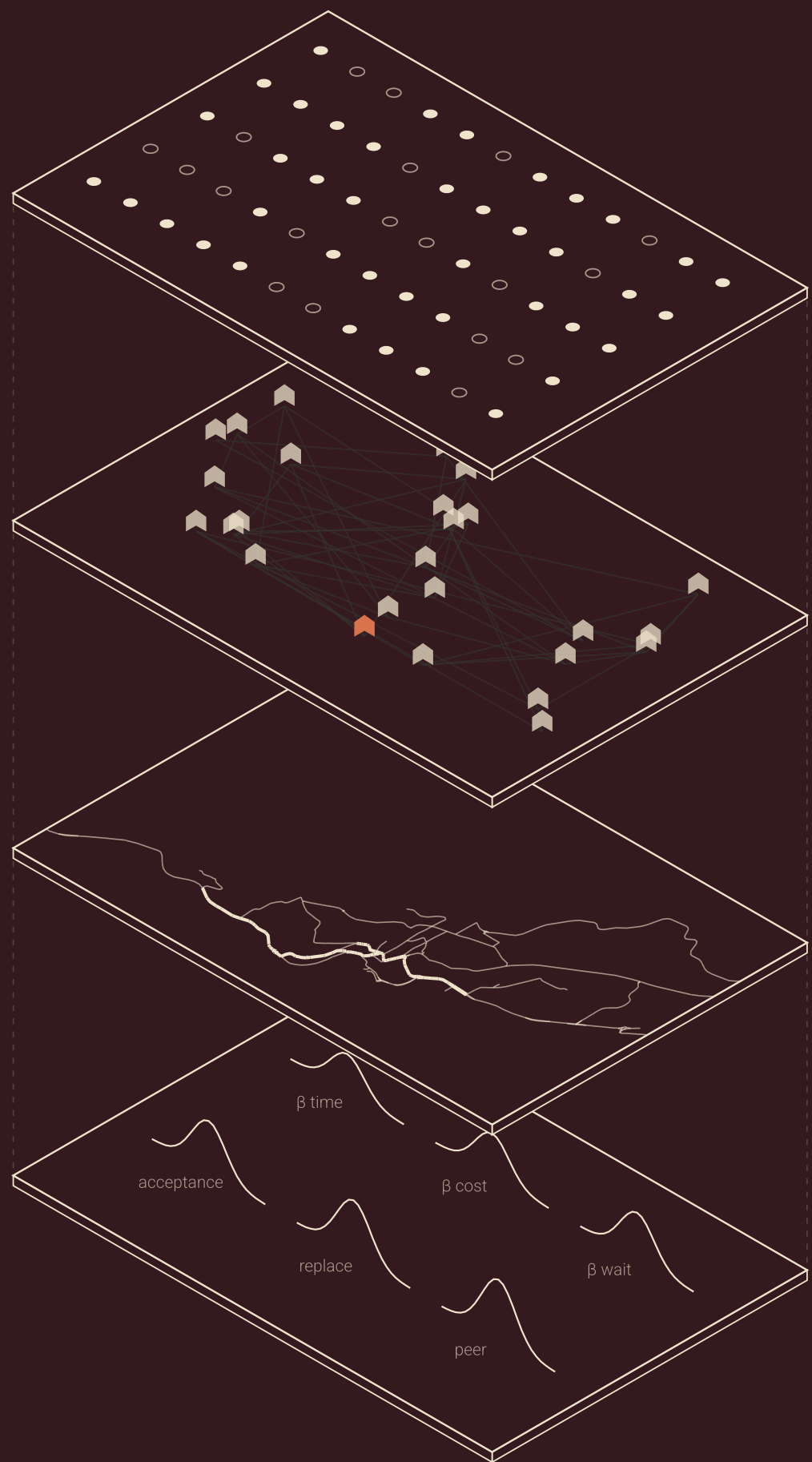
case

LAYER 0

preferences

estimated from people,
with their error

A1



EXPLODED ENGINE

four layers. at the bottom, preferences estimated from people, with their error. above them, the real road network. then households that carry their cars, trips and opinions through time. at the top, the futures every plan is tested against.

LAYER 0 · PREFERENCES · BUILT IN A1

For person n choosing mode j among car, electric car, bus and bike:

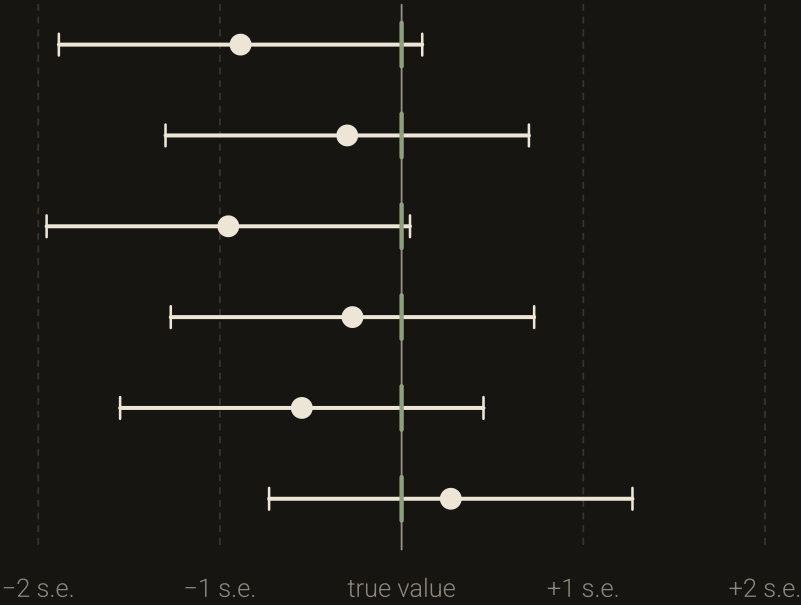
$V_{nj} = ASC_j + \beta_{time} \cdot time_{nj} + \beta_{cost} \cdot cost_{nj} + \beta_{wait} \cdot wait_{nj}, \quad P_{nj} = e^{V_{nj}} / \sum_k e^{V_{nk}}$

Value of time is $60 \beta_{time} / \beta_{cost}$ NOK per hour. In the prototype a synthetic survey of 800 respondents with eight tasks each is simulated from known parameters and estimated twice, with

Biogeme and with an independent maximum-likelihood code. The two agree to 1.5×10^{-6} , and every parameter is recovered within one standard error of its true value. The recovered value of time is 142 NOK per hour against a true 144.

does the estimator find what was put in

electric car constant	-0.848 ± 0.054
bus constant	0.576 ± 0.081
bike constant	-0.477 ± 0.081
time, per minute	-0.0607 ± 0.0024
cost, per NOK	-0.0256 ± 0.0012
wait, per minute	-0.0885 ± 0.0055



SIX OF SIX RECOVERED

each row is one parameter. the tick is the value the synthetic survey was built from, the dot is the estimate, the bar one standard error either side. every bar covers its tick.

docs/results/transplan_study.json; 800 respondents × 8 tasks; Biogeme MNL; synthetic data.

What A1 changes. Real stated-choice data in three blocks (mode, next car, votes on charging packages), mixed and latent-class logit so preferences vary across people, and a hybrid choice model for trust. Each respondent completes two of the

three blocks with four tasks in each, so every block has about 530 respondents, and the sample size follows from the S-estimates of the efficient design, random-parameter spreads included. Stated replacement timing is estimated jointly with revealed replacement from the register.

LAYER 1 · NETWORK · BUILT IN A2

OpenStreetMap road graphs for both towns, filtered to drivable ways, projected, with speeds and travel times on every edge. Paths are computed on the graph, never as straight lines. The

prototype engine still uses three zones with a congestion curve each; A2 moves trips onto the graph with zone data and trip lengths taken from the regional model RTM, so results stay comparable with it.

LAYER 2 · HOUSEHOLDS · BUILT IN A2

Each year every car owner may replace the car. In the prototype that is a constant yearly rate, $1 / \text{lifetime}$; the new car is electric

with probability

$P(\text{EV}) = 1 / (1 + e^{-u}), \quad u = 0.5 - 0.03 \cdot \text{price gap} + 0.08 \cdot \text{running-cost saving} + \varphi \cdot \text{zone EV share}$

where the price gap closes to parity by an uncertain year, the saving includes any toll exemption, and φ is a peer effect. Agents then choose modes with the layer-0 utilities; car times follow a congestion curve solved by successive averages. Welfare is the logsum divided by $-\beta_{\text{cost}}$, and each agent carries the subsidies it receives and the tolls it pays.

LAYER 3 · FUTURES AND PLANS · BUILT IN A3

Nine uncertain inputs span the futures and five levers span the plans. In the prototype both are Latin hypercube samples: 60

UNCERTAIN INPUT	RANGE
EV price parity year	2026 to 2040
Fuel price growth, real	−2% to +4% a year
Telework share of commute days, 2050	0 to 30%
Population growth	0 to 1.5% a year
Peer effect on EV choice	0 to 3
Vehicle lifetime	10 to 18 years
Transfer scale, cost and time sensitivity	0.7 to 1.3 each
Yearly efficiency gain, fuel cars	0 to 3%

A package p is robust in proportion to the futures in which commuting CO₂ in 2050 falls to the target:

$$R(p) = |\{ f \in F : \text{CO}_{22050}(p, f) \leq \tau \}| / |F|, \quad p^* = \operatorname{argmax} R(p) \text{ s.t. } \operatorname{median}_f A(p, f) \geq \tfrac{1}{2}, \quad C(p) \leq B$$

The prototype applies the acceptance constraint only; the budget B will be set with the county from its transport budget. Some prototype levers, such as purchase subsidies and toll exemptions for electric cars, are instruments Norway is now phasing out; A3 replaces them with the instruments debated in 2026, distance-based charging first.

What A2 changes. Replacement depends on car age and household events; the price gap and the purchase model are calibrated to the register so the fleet turns as fast as Norway’s does; acceptance is carried per household and shifts with experience; vehicle and battery production enter as an emission at purchase; repeated runs separate chance from preference error.

futures and 24 packages plus a no-policy reference.

LEVER	RANGE
Toll per car trip	NOK 0 to 60
Share of the toll electric cars pay	0 to 100%
Bus frequency gain	0 to headways halved
Cycle network	none to complete
EV purchase subsidy	NOK 0 to 100,000

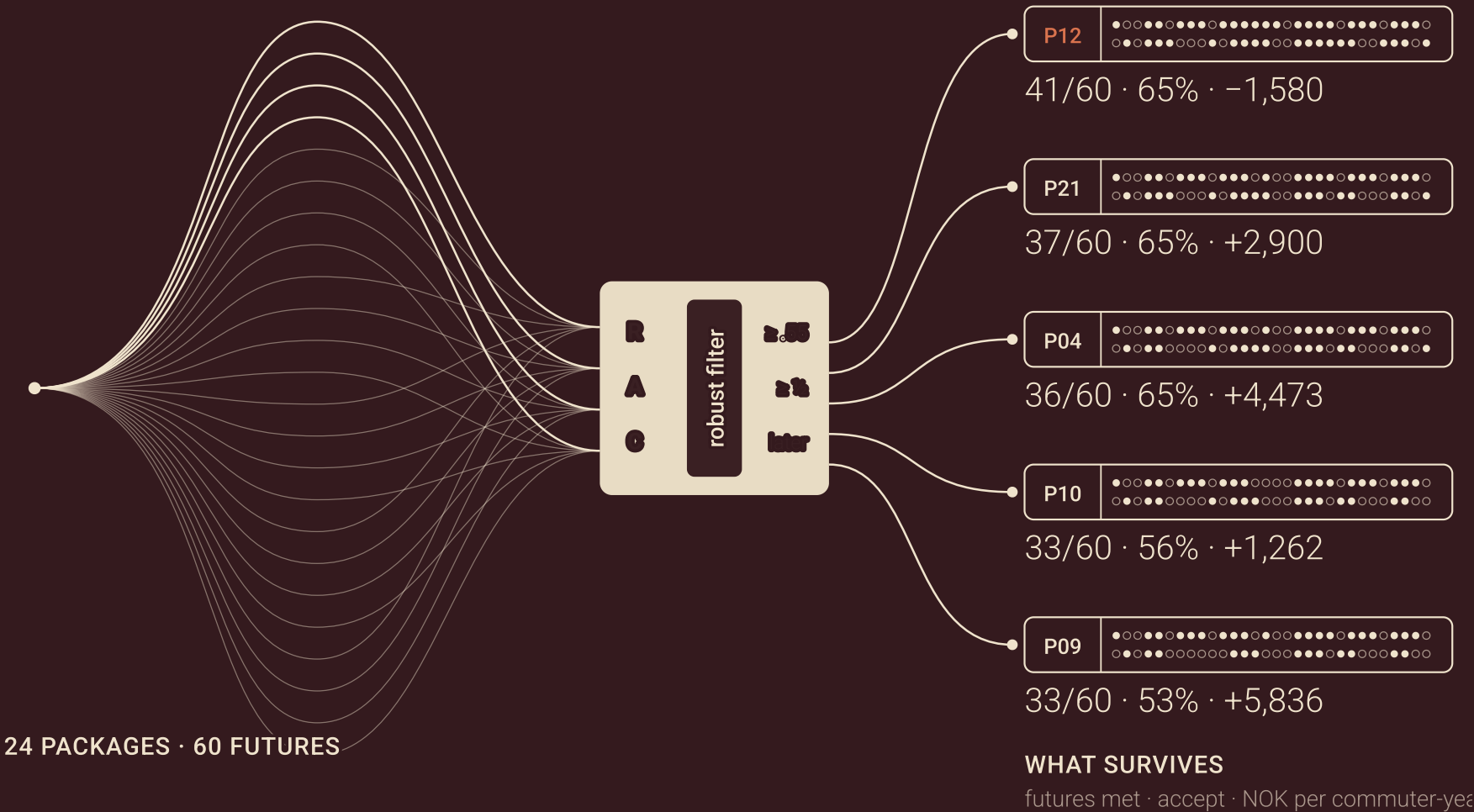
What A3 changes. About 10⁴ futures anchored on the national strategies¹⁶, packages screened as a 10² Latin hypercube and then refined by a many-objective search in the best region¹⁹, τ set against the Climate Change Act, which implies about 5 to 10% of 2025 rather than the prototype’s looser 20%, reported at 5, 10 and 20%, regret and stability beside robustness²⁰, and a run budget: one prototype run takes about 0.1 s with 1,000 agents, so the full design is costed before it is launched and a surrogate is fitted if one full-scale run passes ten seconds.

WHAT THE PROTOTYPE FOUND

1,500 runs: 24 packages and a no-policy reference, each in the same 60 futures, on synthetic behaviour. The numbers show what each

article will produce, not what will happen in Norway. Four results already shape the plan.

selective survival

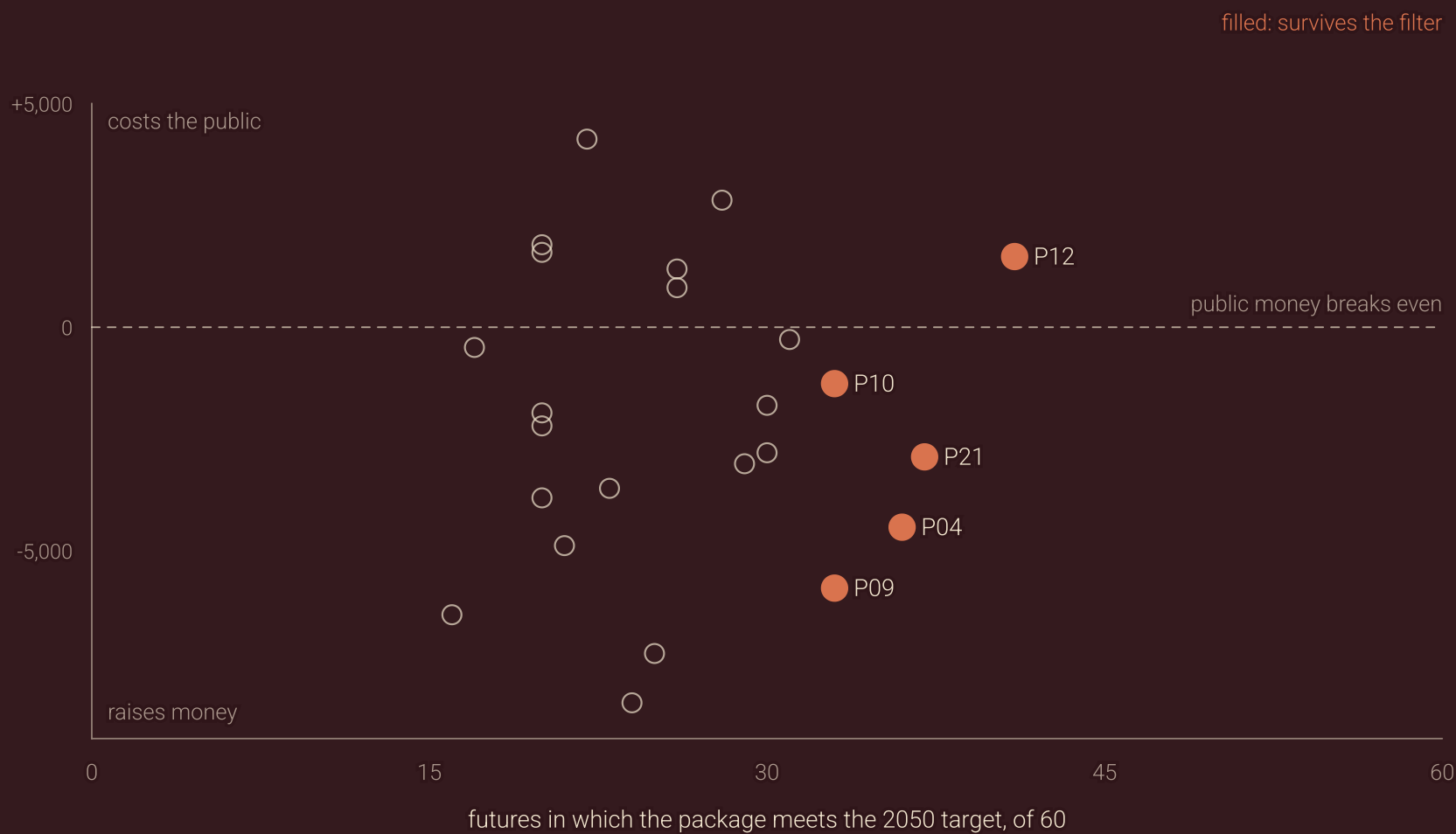


FIVE PLANS SURVIVE

a package survives if it meets the target in at least 55% of futures and the median future leaves a majority not worse off. each dot is one future, lit where the package met the target. the best, P12, meets it in 41 of 60.

Prototype ensemble; target: commuting CO₂ in 2050 at most 20% of 2025; acceptance is the self-interest proxy; budget not yet applied.

what robustness costs the public purse

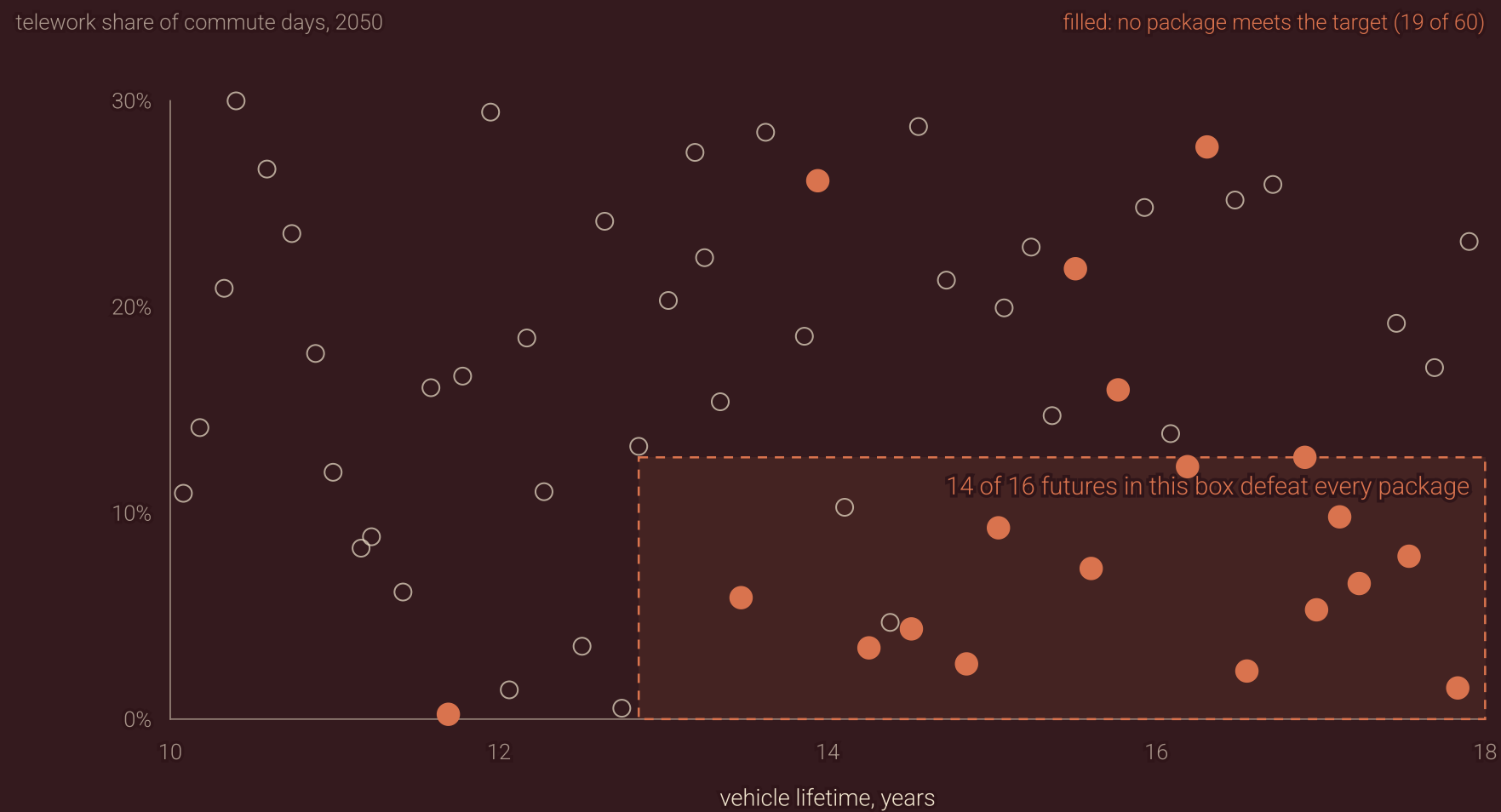


THE PRICE OF ROBUSTNESS

each circle is a package: across, the futures in which it meets the target; up, the public money it moves per commuter and year, subsidies paid minus tolls collected. four of the five survivors raise money through tolls; the most robust, P12, costs about NOK 1,580 per commuter-year. the packages that cost the most are not the robust ones.

Prototype ensemble; medians over 60 futures; NOK per commuter per year, 2025 to 2050 average.

where every plan fails

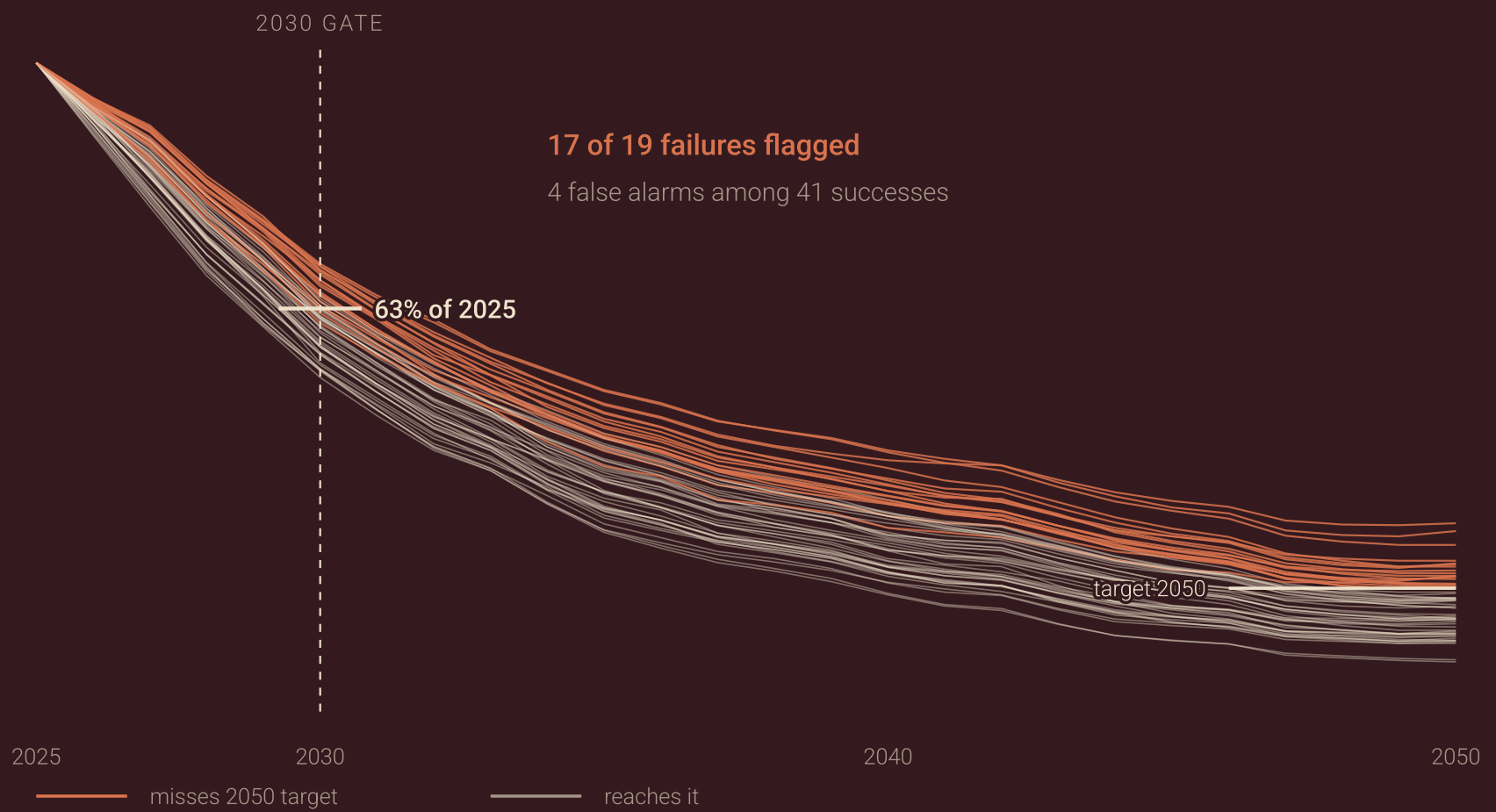


WHEN CARS LAST

in 19 of the 60 futures no package meets the target. 14 of them sit in one box: cars that last 12.9 years or more and telework that stays under 13% of commute days. the policy question there is not which package, but how to shorten the life of the fleet.

Prototype ensemble; exhaustive two-threshold search, 74% of hostile futures inside, 88% of futures inside are hostile.

reading the future early



A WARNING WITH A HIT RATE

sixty futures under P12. at the 2030 gate, emissions above 63% of 2025 flag 17 of the 19 futures that later miss the target, with 4 false alarms among 41 that succeed: a warning in 2030, twenty years before the target year. the threshold was fitted on these futures; A3 scores it on held-out ones.

Prototype ensemble; threshold chosen for the best separation of later success and failure.

THE TOOLBENCH

Every result above comes from a tool in the repository, each with its own tests. The bench is laid out the way it is used: the call, then what it

returned. The second half lists what is not built yet, which article builds it, and why it is not faked in the meantime.

DCM-ESTIMATE

What do people trade off?

```
sp = dcm.simulate(800, 8, seed=1)
bio = dcm.estimate_biogeme(sp)
mle = dcm.estimate_mle(sp)
```

6 of 6 parameters within one standard error; Biogeme and own MLE agree to 1.5×10^{-6} ; value of time 142 NOK/h.

5 tests

ABM-TRANSPORT

How does the fleet turn?

```
out = abm.run(levers={"toll_nok": 30},
               uncertainties=
               {"ev_parity_year": 2030},
               beta=bio["beta"],
               n=1000)
```

26 years of mode shares, EV fleet share, CO₂, consumer surplus, subsidies and tolls per commuter, in about 0.1 s.

10 tests

DMDU-EXPLORE

Which plans survive which futures?

```
exp = dmdu.explore(60, 24, 1000, beta=beta)
rob = dmdu.robustness(exp, 0.2)
box = dmdu.discover(exp, "P17", 0.2)
```

1,500 runs with EMA Workbench; robustness per package, PRIM boxes, backcast milestones.

6 tests

OSM-NETWORK

Where does the network concentrate travel?

```
G =
osm_network.build(place="Molde, Norway")
p = osm_network.route(G, a, b,
                       weight="travel_time")
```

2,772 junctions in Molde and 3,066 in Kristiansund, with speeds, times and energy per edge.

9 tests

VRP-SOLVE

How should a van drive the town?

```
res = vrp_solve.solve(G, problem,
                      solver="pyvrp")
# cross-check: solver="ortools"
```

48 min instead of 137 for 24 addresses; two independent solvers within 0.1%.

4 tests

OPT-MODEL

Exact models when size allows

```
m = models.facility_location(
    fixed, capacity, demand,
    cost)
solvers.solve(m) # Gurobi, CPLEX or HiGHS
```

Transportation, facility location, min-cost flow and exact CVRP; free solver limits measured (Gurobi 2,000 variables).

8 tests

VIZ-STORY

Can a planner see the pathways?

```
python story.py --out docs/story \
  --futures 30 --agents 600
```

SimWrapper dashboard of pathway bands, a kepler.gl map of every agent's commute, and a data tree for Grasshopper.

5 tests

VISUAL-NARRATIVE

One idea per plate

```
python lineart/render.py stock-
turns \
  out.png charcoal
```

The line-art grammar behind every plate in this portfolio, drawn from the study data, never by hand.

4 tests

Not built yet, and not faked

SURVEY DESIGN

A1 · month 2 to 6

Efficient design with S-estimates for all three blocks, priors from the pilot. Until then the prototype uses a synthetic survey and says so.

HYBRID CHOICE MODEL

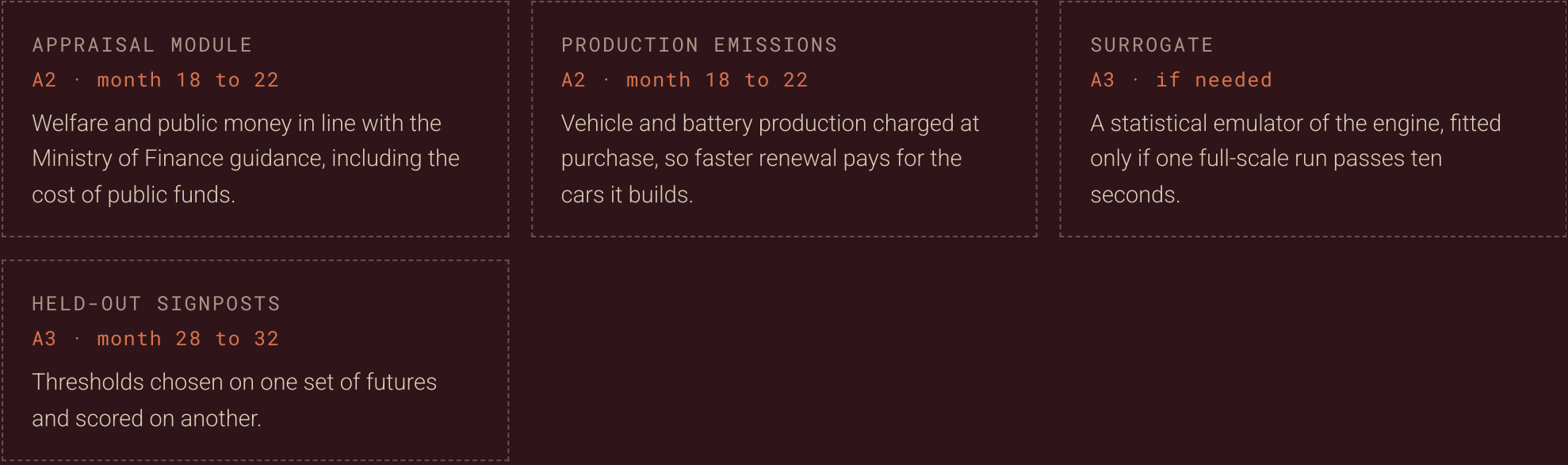
A1 · month 8 to 12

Trust and attitudes as latent variables measured by indicators, feeding acceptance.

REGISTER CALIBRATION

A2 · month 13 to 18

Purchase and replacement fitted to vehicle register extracts at zone level, and checked against BIG.



RERUN IT YOURSELF

No number in the proposal or this portfolio needs to be taken on trust. Four commands rebuild them from a clean checkout, and the tests run

1

```
git clone https://github.com/Astheno-sphere/computational_Logistics
cd computational_Logistics
```

The code, the data scripts and the documents live together.

3

```
python examples/transplan_study.py
python docs/proposal/make_data.py
```

The survey, the estimation, the 1,500 runs, the robustness table and every plate number. Asserts stop the build if the ensemble drifts.

The repository separates what runs, what is tested and what is planned, keeps a claims policy that ties every number to a script or a

on every change.

2

```
pip install -r requirements-lock.txt
```

Exact package versions, so a rerun next year gives the same numbers.

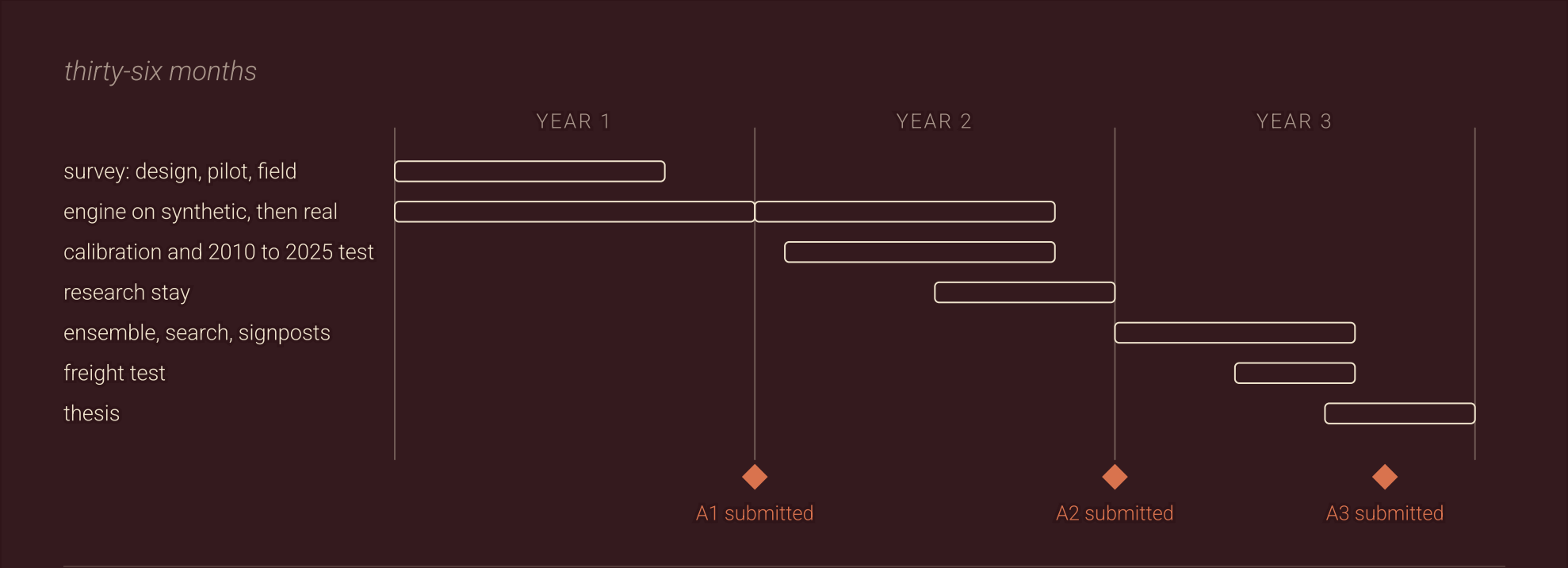
4

```
python -m pytest -q tests
```

87 tests across the tools; the same suite runs in continuous integration on every push.

source, and lists its limitations on the front page.

THREE YEARS, AND THE LIMITS



ONE ARTICLE A YEAR

the engine is built in year 1 on synthetic preferences while the survey is in the field, so year 2 swaps in real preferences rather than starting from zero. each year ends with a submitted article.

If the survey slips. If fieldwork runs past month 9, A1 is submitted on the pilot and panel sample, and the postal wave feeds A2. Recruitment combines a Norwegian web panel with postal invitations drawn from the population register, so coverage does not depend on one channel.

If data access slips. Statistics Norway commuting tables stand in for the travel survey; register extracts are requested from the Norwegian

Public Roads Administration in month 1, at zone level, so no individual records are needed.

If run time explodes. The design is costed from measured run times before launch, and a surrogate is fitted if needed, on national computing resources.

What this work does not do

- Every behavioural number in this portfolio is synthetic until A1. The prototype shows the chain works, not what Norwegians prefer.
- The prototype fleet turns more slowly than Norway's and is not yet calibrated; its EV shares are not forecasts.
- The prototype applies acceptance as a self-interest proxy and does not yet apply the budget constraint.
- The prototype target, commuting CO₂ in 2050 at most 20% of 2025, is looser than the 5 to 10% the Climate Change Act implies; every prototype number would be harder to reach under the law.
- Scope is commuting and other regular car travel by residents of two towns, translated to national targets; it is not a national model and does not replace NTM6 or RTM.
- The freight plates use free-flow times without congestion or turn penalties; the freight test in A3 is a test of the method, not a freight demand model.

The fleet that will drive in 2035 is being bought now. I have built the first version of the instrument to steer it. With TRANSPLAN's data, Molde's supervision and three years, it becomes one that planners can trust.

1. OFV (2026) New passenger car registrations, 2025.
2. Kristensen et al. (2024) *Transport demand in foresight and backcasting*. TØI report 2051/2024.
3. Ministry of Transport (2024) Meld. St. 14 (2023–2024) *National Transport Plan 2025–2036*.
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5. Wangsness, Proost & Steinsland (2026) The curious case of road pricing reform in Norway. *Transportation Research Interdisciplinary Perspectives* 38, 102064.
6. Fridstrøm, Østli & Johansen (2016) A stock-flow cohort model of the national car fleet. *European Transport Research Review* 8.
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8. Grimm et al. (2005) Pattern-oriented modeling of agent-based complex systems. *Science* 310.
9. Le Pira, Marcucci, Gatta, Inturri, Ignaccolo & Pluchino (2017) *Research in Transportation Economics* 64; Gatta et al. (2020) *Transportation Research Procedia* 46.
10. Ciccone, Halse, Andreassen, Garnache & Wangsness (2026) *Acceptance of distance-based road pricing: national choice experiment and field trial in the Oslo region*. TØI report 2179/2026.
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13. Tveter, Welde & Odeck (2025) Accounting for uncertainties in cost-benefit analyses of road projects. *Transport Policy* 170.
14. Raso, Kwakkel & Timmermans (2019) Assessing the capacity of adaptive policy pathways to adapt on time. *Sustainability* 11(6), 1716.
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16. Wangsness et al. (2025) *Methods, processes and example calculations for strategies towards a carbon-neutral transport sector in 2050*. TØI report 2119/2025.
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18. Welde, Tveter & Odeck (2019) The traffic effects of fixed links: short and long-run forecast accuracy. *Transportation Research Procedia* 42.
19. Kasprzyk, Nataraj, Reed & Lempert (2013) *Environmental Modelling & Software* 42; Kwakkel (2017) *Environmental Modelling & Software* 96.
20. Shaabani, Hvattum, Laporte & Hoff (2024) Stability metrics for a maritime inventory routing problem under sailing time uncertainty. *EURO Journal on Transportation and Logistics* 13.